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FINAL YEAR PROJECT · PYTHON · MACHINE LEARNING · DESKTOP APPLICATION

Smart System Performance Dashboard

A real-time hardware monitoring and intelligent fault detection application for Windows

Python | PyQt6 | scikit-learn | SQLite | pyqtgraph | psutil / pynvml

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01 Overview

Tools like Windows Task Manager show technical numbers and graphs, but do not explain what they mean or whether anything is actually wrong.

This project aims to fix that. It is a desktop dashboard that watches your computer's hardware in real-time and uses machine learning to spot problems automatically, telling you what is wrong in plain, simple language without needing any technical knowledge.

02 How It Works

Four independent services run in parallel, each responsible for a different stage of the pipeline:

Collector Service
CPU, RAM, GPU & Disk metrics sampled every 1 s via Windows APIs

Storage Service
SQLite database storing all telemetry data per session

Analytics Service
Runs the Random Forest model every 5 s on a 10 s window of telemetry

Dashboard Service
Live PyQt6 GUI with real-time graphs, health score & issue detection panel

03 Machine Learning Pipeline

A Random Forest classifier is trained to detect 12 categories of hardware issue. Training labels are generated by applying 12 hand-written threshold rules to real and synthetic system telemetry. The resulting 200-tree model runs every 5 seconds over a rolling 10-second window of telemetry data.

20,000+	0.98	42
TRAINING SAMPLES	AVG F1 SCORE	FEATURES

04 Detected Issues

The model can detect 12 hardware issues across four components:

COMPONENT	DETECTED ISSUES
CPU	Thermal Throttle · Bottleneck · Sustained High Load
RAM	Memory Pressure · Memory Leak · Excessive Swap
DISK	Disk Full · Bottleneck · High Latency
GPU	Overheating · Power Throttle · VRAM Pressure

05 Technologies Used

The application is built entirely in Python, using the following libraries and frameworks:

Python | PyQt6 | scikit-learn | pyqtgraph | pandas | numpy | psutil | pynvml | pywin32 | SQLite | pytest

06 Testing

Over 80 unit & other tests across three suites. Edge cases include missing GPU hardware, permission-denied disk partitions, Windows 10/11 PDH frequency fallback, and boundary conditions for all 12 detection rules.

PASS	PASS	PASS
COLLECTOR	STORAGE	ANALYTICS

07 System Architecture

DATA SOURCES — OS / Hardware Layer
- psutil — CPU, RAM, Disk
- pynvml — GPU metrics
- pywin32 — PDH counters

→ API calls

COLLECTOR SERVICE — QThread · 1 s loop
- Polls CPU, RAM, GPU & Disk every 1 s
- Collects system info on startup
- Writes each sample to the database

↓ 1 s write

STORAGE SERVICE — SQLite · WAL Mode
- Normalised schema — 11 tables
- Per-session & per-machine tagging
- Supports reading and writing at the same time

← 5 s read

ANALYTICS SERVICE — ML Inference · 5 s cycle
- Multi-output Random Forest classifier
- 42 features over 10-sample window
- Detects 12 issues across 4 components

↓ insights

↓ 1 s live data

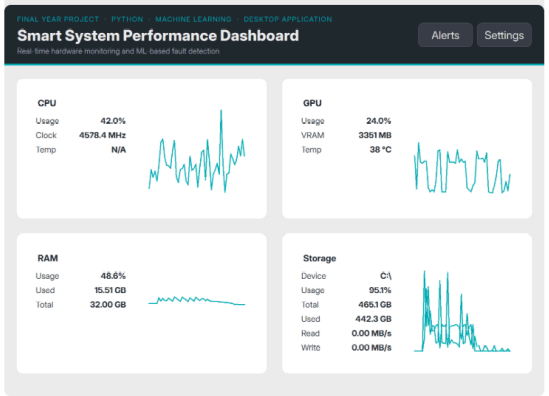
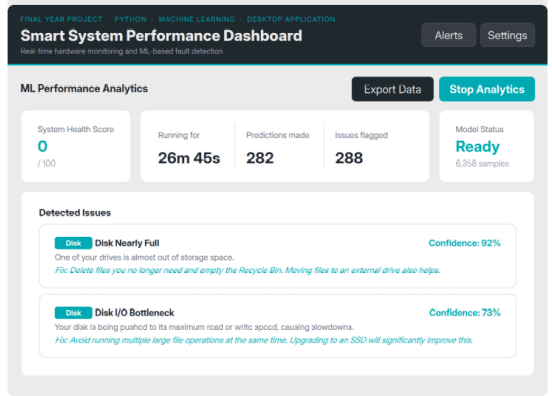
DASHBOARD SERVICE — PyQt6 · GUI Frontend

LIVE MONITOR	ANALYTICS PANEL	GENERAL UI
Real-time scrolling graphs — CPU, RAM, GPU, Disk	System Health Score (0–100)	Tab navigation · Settings · CSV export
2×2 pyqtgraph grid with hover tooltips	Lists detected issues in plain language with confidence %	Alert button for detected issues

08 Results

The screenshots below show the application running on a live system. The left shows the real-time monitor with scrolling graphs for CPU, RAM, GPU and disk. The right shows the analytics panel mid-session, with a detected issue, component risk bars and a system health score.

LIVE SYSTEM MONITOR



ANALYTICS PANEL